

Compare Permissions on Regular Files and Directories

Create a regular file named sample.txt and a directory named sample_dir. Create at least one file inside sample_dir.

Apply each of the following permissions using chmod. Perform the specified operations and record your observations.

Note: Perform the exercise as a normal user. Do not use sudo.

Permission	Regular File (sample.txt) – Test	Observation	Directory (sample_dir) – Test	Observation
400(r--)	Try to read and modify the file	Will be able to read successfully but can't modify it(permission denied)	Try to list and enter the directory	Will list the filename but can't access it and cannot enter the directory
200(-w-)	Try to read and modify the file	We cannot read the file but can modify it	Try to list, enter,create a file	Cannot list,enter or create a file
100(--x)	Try to read and execute the file	Cannot read or execute the file	Try to list and enter the directory	Cannot list the directory but we can enter it.
500(r-x)	Try to read and modify the file	Will be able to read the file but cannot modify it	Try to list and enter the directory	Can list and enter the directory
600(rw-)	Try to read and modify the file	Can both read and modify the file	Try to list,enter and create a file	Can list filename but cannot enter directory or create files
700(rwx)	Try to read, modify and execute the file	Will be able to read, modify and execute the file	Try to list,enter , create and delete files	Will be able to list,enter, create and delete files

```
kali@kali: ~  
Session Actions Edit View Help  
  
(kali@kali)-[~]  
$ touch sample.txt  
  
(kali@kali)-[~]  
$ mkdir sample_dir  
  
(kali@kali)-[~]  
$ touch sample_dir/file1.txt  
  
(kali@kali)-[~]  
$ ls -l  
total 52  
drwxr-xr-x 2 kali kali 4096 Aug  8 11:43 Desktop  
drwxr-xr-x 2 kali kali 4096 Aug 15 11:08 Documents  
drwxr-xr-x 2 kali kali 4096 Aug 15 11:03 Downloads  
drwxrwxr-x 2 kali kali 4096 Aug 10 12:30 example  
drwxrwxr-x 3 kali kali 4096 Aug 10 15:11 LinuxOS_2026  
drwxr-xr-x 2 kali kali 4096 Aug  8 11:43 Music  
drwxrwxr-x 3 kali kali 4096 Aug 11 07:05 phishing_simulator  
drwxr-xr-x 2 kali kali 4096 Aug  8 11:43 Pictures  
drwxr-xr-x 2 kali kali 4096 Aug  8 11:43 Projects  
drwxr-xr-x 2 kali kali 4096 Aug  8 11:43 Public  
drwxrwxr-x 2 kali kali 4096 Aug 15 11:17 sample_dir  
-rw-rw-r-- 1 kali kali    0 Aug 15 11:17 sample.txt  
drwxr-xr-x 2 kali kali 4096 Aug  8 11:43 Templates  
drwxr-xr-x 2 kali kali 4096 Aug  8 11:43 Videos
```

400(r--)

```
(kali@kali)-[~]  
$ chmod 400 sample.txt  
  
(kali@kali)-[~]  
$ chmod 400 sample_dir  
  
(kali@kali)-[~]  
$ ls -ld sample.txt sample_dir  
dr----- 2 kali kali 4096 Aug 15 11:17 sample_dir  
-r----- 1 kali kali    0 Aug 15 11:17 sample.txt
```

chmod 400 sample.txt

```
(kali@kali)-[~]  
$ cat sample.txt  
  
(kali@kali)-[~]  
$ echo "Hello" > sample.txt  
zsh: permission denied: sample.txt
```

chmod 400 sample_dir

```
(kali@kali)-[~]  
$ ls sample_dir  
ls: cannot access 'sample_dir/file1.txt': Permission denied  
file1.txt  
  
(kali@kali)-[~]  
$ cd sample_dir  
cd: permission denied: sample_dir
```

200(-w-)

```
(kali㉿kali)-[~]
└─$ chmod 200 sample.txt

(kali㉿kali)-[~]
└─$ chmod 200 sample_dir

(kali㉿kali)-[~]
└─$ ls -ld sample.txt sample_dir
d-w----- 2 kali kali 4096 Aug 15 11:17 sample_dir
--w----- 1 kali kali   0 Aug 15 11:17 sample.txt

(kali㉿kali)-[~]
└─$
```

chmod 200 sample.txt

```
(kali㉿kali)-[~]
└─$ cat sample.txt
cat: sample.txt: Permission denied

(kali㉿kali)-[~]
└─$ echo "Hello" > sample.txt

(kali㉿kali)-[~]
└─$
```

chmod 200 sample_dir

```
(kali㉿kali)-[~]
└─$ ls sample_dir
ls: cannot open directory 'sample_dir': Permission denied

(kali㉿kali)-[~]
└─$ cd sample_dir
cd: permission denied: sample_dir

(kali㉿kali)-[~]
└─$ touch sample_dir/file2.txt
touch: cannot touch 'sample_dir/file2.txt': Permission denied

(kali㉿kali)-[~]
└─$
```

100(--x)

```
(kali㉿kali)-[~]
└─$ chmod 100 sample.txt

(kali㉿kali)-[~]
└─$ chmod 100 sample_dir

(kali㉿kali)-[~]
└─$ ls -ld sample.txt sample_dir
d--x----- 2 kali kali 4096 Aug 15 11:17 sample_dir
--x----- 1 kali kali   6 Aug 15 12:32 sample.txt

(kali㉿kali)-[~]
└─$
```

Chmod 100 sample.txt

```
(kali㉿kali)-[~]
$ cat sample.txt
cat: sample.txt: Permission denied

(kali㉿kali)-[~]
$ ./sample.txt
zsh: permission denied: ./sample.txt

(kali㉿kali)-[~]
$
```

chmod 100 sample_dir

```
(kali㉿kali)-[~]
$ ls sample_dir
ls: cannot open directory 'sample_dir': Permission denied

(kali㉿kali)-[~]
$ cd sample_dir

(kali㉿kali)-[~/sample_dir]
$ pwd
/home/kali/sample_dir
```

500(r-x)

```
(kali㉿kali)-[~]
$ chmod 500 sample.txt

(kali㉿kali)-[~]
$ chmod 500 sample_dir

(kali㉿kali)-[~]
$ ls -ld sample.txt sample_dir
dr-x----- 2 kali kali 4096 Aug 15 11:17 sample_dir
-r-x----- 1 kali kali    6 Aug 15 12:32 sample.txt

(kali㉿kali)-[~]
$
```

chmod 500 sample.txt

```
(kali㉿kali)-[~]
$ cat sample.txt
Hello

(kali㉿kali)-[~]
$ echo "Hi" > sample.txt
zsh: permission denied: sample.txt
```

chmod 500 sample_dir

```
(kali㉿kali)-[~]
└─$ ls sample_dir
file1.txt

(kali㉿kali)-[~]
└─$ cd sample_dir

(kali㉿kali)-[~/sample_dir]
└─$ pwd
/home/kali/sample_dir

(kali㉿kali)-[~/sample_dir]
└─$
```

600(rw-)

```
(kali㉿kali)-[~]
└─$ chmod 600 sample.txt

(kali㉿kali)-[~]
└─$ chmod 600 sample_dir

(kali㉿kali)-[~]
└─$ ls -ld sample.txt sample_dir
drw----- 2 kali kali 4096 Aug 15 11:17 sample_dir
-rw----- 1 kali kali   6 Aug 15 12:32 sample.txt
```

chmod 600 sample.txt

```
(kali㉿kali)-[~]
└─$ cat sample.txt
Hello

(kali㉿kali)-[~]
└─$ echo "Goodmorning" > sample.txt

(kali㉿kali)-[~]
└─$ cat sample.txt
Goodmorning
```

chmod 600 sample_dir

```
(kali㉿kali)-[~]
└─$ ls sample_dir
ls: cannot access 'sample_dir/file1.txt': Permission denied
file1.txt

(kali㉿kali)-[~]
└─$ cd sample_dir
cd: permission denied: sample_dir

(kali㉿kali)-[~]
└─$ touch sample_dir/file3.txt
touch: cannot touch 'sample_dir/file3.txt': Permission denied
```

700(rwx)

```
(kali㉿kali)-[~]
└─$ chmod 700 sample.txt

(kali㉿kali)-[~]
└─$ chmod 700 sample_dir

(kali㉿kali)-[~]
└─$ ls -ld sample.txt sample_dir
drwx----- 2 kali kali 4096 Aug 15 11:17 sample_dir
-rwx----- 1 kali kali  12 Aug 15 12:44 sample.txt
```

chmod 700 sample.txt

```
(kali㉿kali)-[~]
└─$ cat sample.txt
Goodmorning

(kali㉿kali)-[~]
└─$ echo "Goodnight" > sample.txt

(kali㉿kali)-[~]
└─$ cat sample.txt
Goodnight

(kali㉿kali)-[~]
└─$ ./sample.txt
./sample.txt: 1: Goodnight: not found
```

chmod 700 sample_dir

```
(kali㉿kali)-[~]
└─$ cat sample.txt
Goodnight

(kali㉿kali)-[~]
└─$ ./sample.txt
./sample.txt: 1: Goodnight: not found

(kali㉿kali)-[~]
└─$ cd sample_dir

(kali㉿kali)-[~/sample_dir]
└─$ touch new.txt

(kali㉿kali)-[~/sample_dir]
└─$ ls
file1.txt  new.txt

(kali㉿kali)-[~/sample_dir]
└─$ rm new.txt

(kali㉿kali)-[~/sample_dir]
└─$ ls
file1.txt
```

Questions

What does read (r) permission allow you to do with a regular file?

- It allows to view or copy the file contents

What does read (r) permission allow you to do with a directory?

- It is used to list the name of all the files and folders inside the directory

What does write (w) permission mean for a regular file?

- It allows permission to modify and make changes to the file

What does write (w) permission mean for a directory?

- It allows to create, delete or edit files in the directory

What does execute (x) permission mean for a regular file?

- It allows the run the file as a program or a script

Why is execute (x) permission important for accessing a directory?

- Execute permission will allow us to enter the directory and access the contents inside it.

Which permission combination allows the owner to fully access a regular file?

- 700 that is rwx-----

Which permission combination allows the owner to fully access a directory?

- 700 that is rwx-----

Based on your observations, explain why the same r, w, and x permissions behave differently for a regular file and a directory.

- The functionalities of the permissions are different for regular file and directory. In a regular file r is used for reading the content, w is used for editing and x is used for executing it. Whereas in a directory r is used to list its contents, w is used to create or delete entries and x is used to enter and access things inside the directory